

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II
SOFTWARE ENGINEERING – LECTURE 16

Virtualization in the Deployment of Complex Systems

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Virtualization

- Virtualization is a technique for increasing the effectiveness and efficiency of complex data centers.
- This technique dates back to the 1960s, but came back into vogue about ten years ago thanks to technological advances in CPUs.

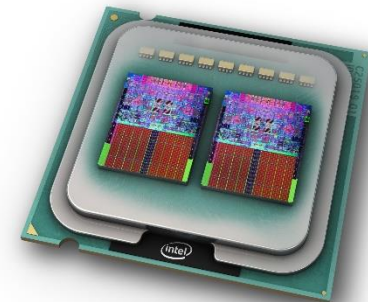
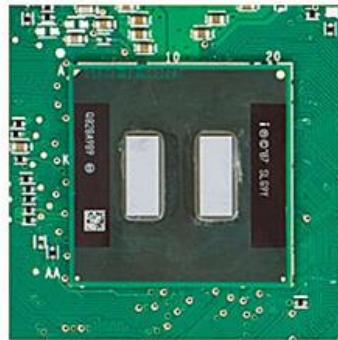
The evolution of CPUs

Evolution of CPUs

- Microprocessors have always seen rapid increases in performance and decreases in cost.
- The strategy for improving performance has historically been to increase the “speed” of a processor.
 - 1978: Intel 8086, 4MHz
 - 1985: Intel 386, 25 MHz (~ 6x)
 - 1995: Intel Pentium Pro, 200 MHz (~ 8x)
 - 2005: Intel Pentium P4 3.8 GHz (3800 MHz) (~ 19x)
- This strategy came to a halt in 2005.
 - 2020: Intel i9-10980XE 3.0 GHz

Multicore CPUs

- Having reached the physical limits of silicon, the solution was to increase the number of processing units (multi-core) in order to increase computing power.
 - 2020: Intel i9-10980XE 3,0 GHz → 18 computing units
 - 2020: AMD Ryzen™ Threadripper™ 3970X 3,7 GHz → 32 computing units



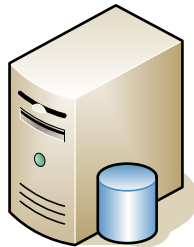
Multi-core issues

- Traditionally, programs have been written to run on a computer with a single CPU.
- The vast majority of applications consist of sequential programs.
- How can multiple units be used in parallel?

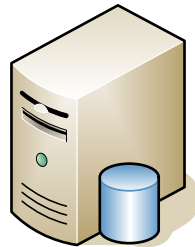
Traditional data center infrastructure

Traditional data center Infrastructure

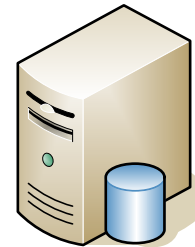
- Individual servers offering individual services and/or applications (one workload, one box)
- New Service and/or Application → New Server, with its own Storage



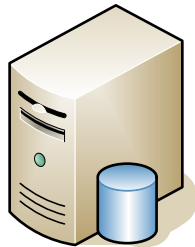
Server Application



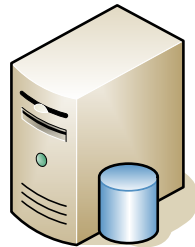
Server di Dominio



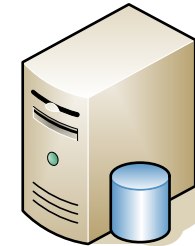
Server Database



Server di Posta



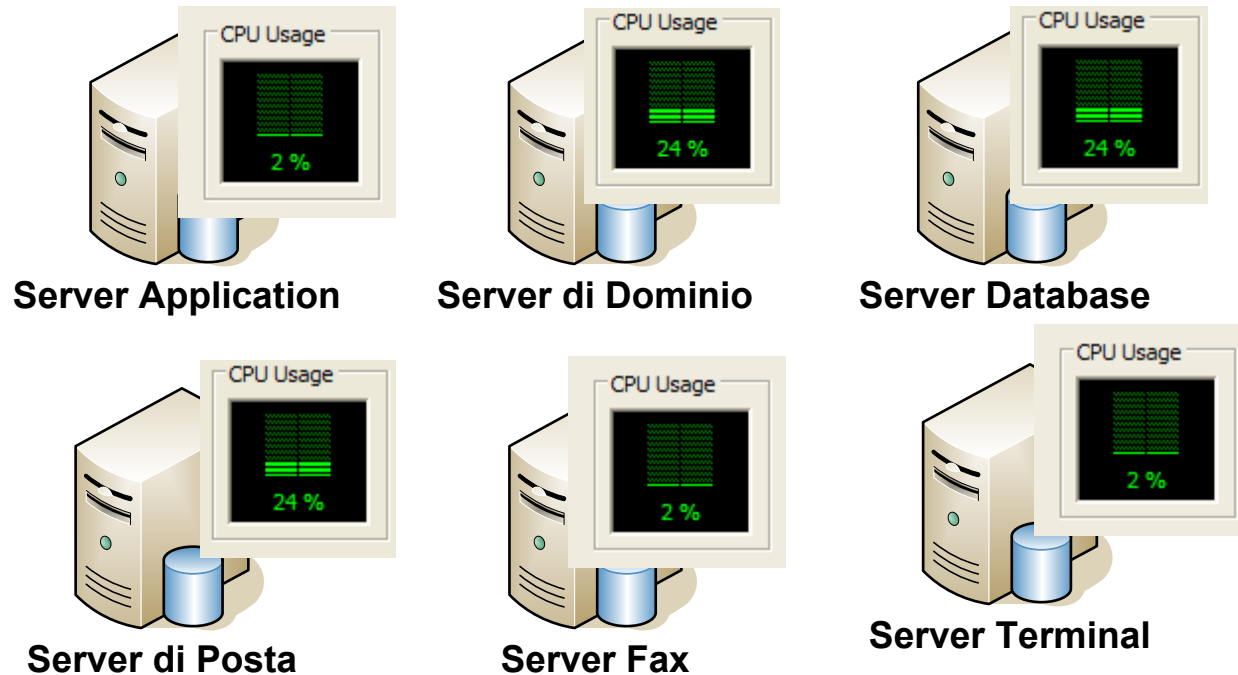
Server Fax



Server Terminal

Traditional data center Infrastructure

- All servers consume (a lot of) electricity, emit heat, take up space, and need to be physically managed...

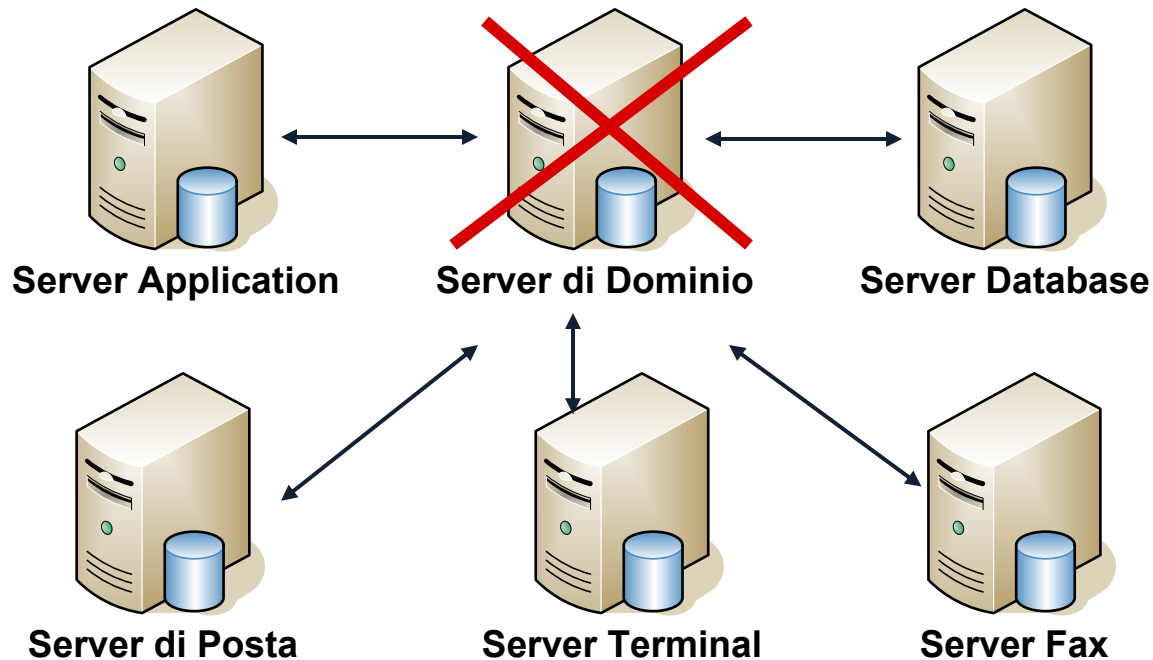


Traditional data center Infrastructure: Consequences

- Underutilization of resources
 - Most servers use between 5% and 15% of their computing capacity on average.
 - In the case of critical services, redundancy is required (backup computers that are not normally used).
- Increased installation and management costs
 - Very expensive nationwide updates
 - Backups must be performed on every machine
 - ...

Traditional data center Infrastructure: Consequences

- The services are intrinsically interconnected.
 - E.g.: The domain server, which manages user authentication, is queried by all other services/servers.
 - What happens if a server breaks down?



Server failure in a traditional scenario

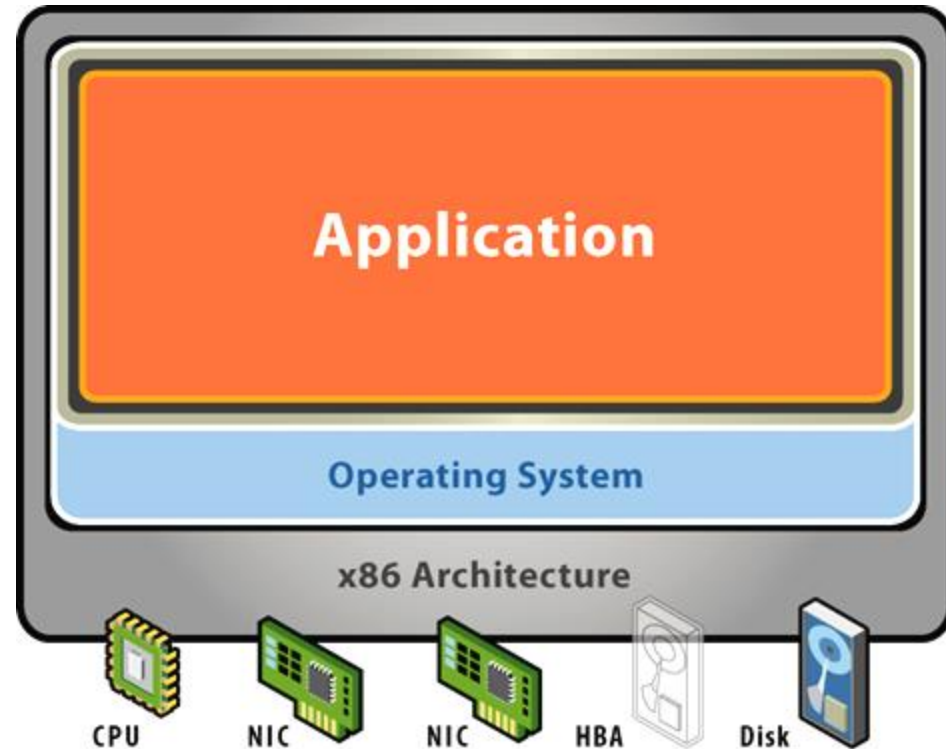
- If one server fails, it can block the services of other servers that communicate with it.
- This results in:
 - Interruption of operations linked to that service.
 - In the most serious cases, interruption of all company operations.
 - Increased costs (downtime and recovery costs).
 - Downtime that is not always predictable.

Virtual Machines

General Concepts

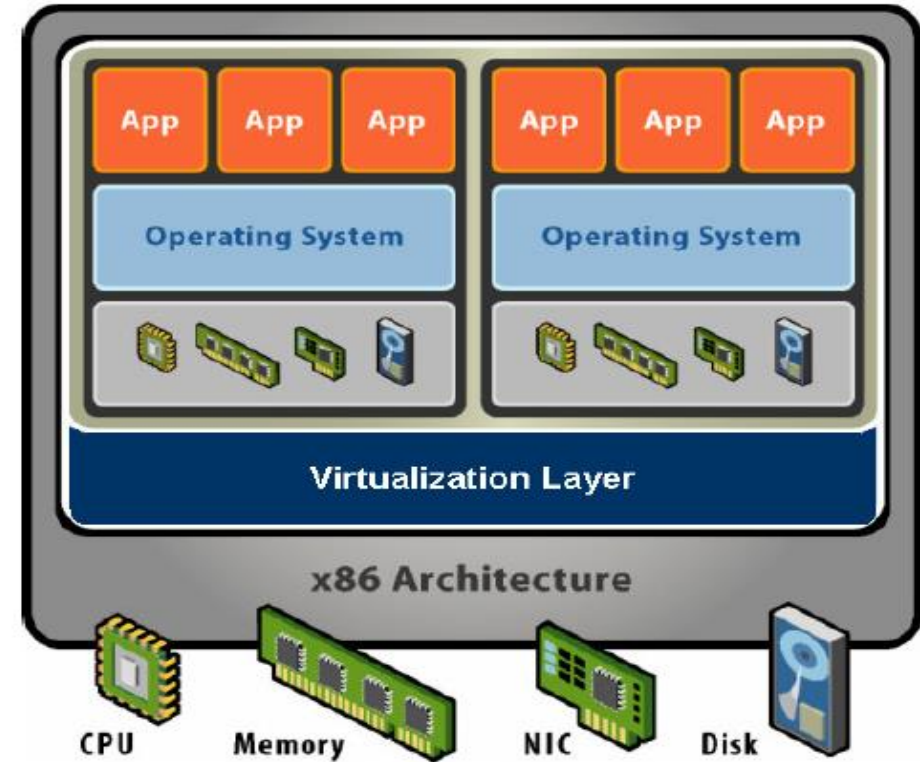
A Physical Server

- A physical server consists of:
 - A series of hardware resources
 - An operating system
 - One or more application software programs
- **Idea:** if the server CPU has multiple parallel computing units, can we “simulate” the existence of multiple computers, independent of the underlying “real” hardware?



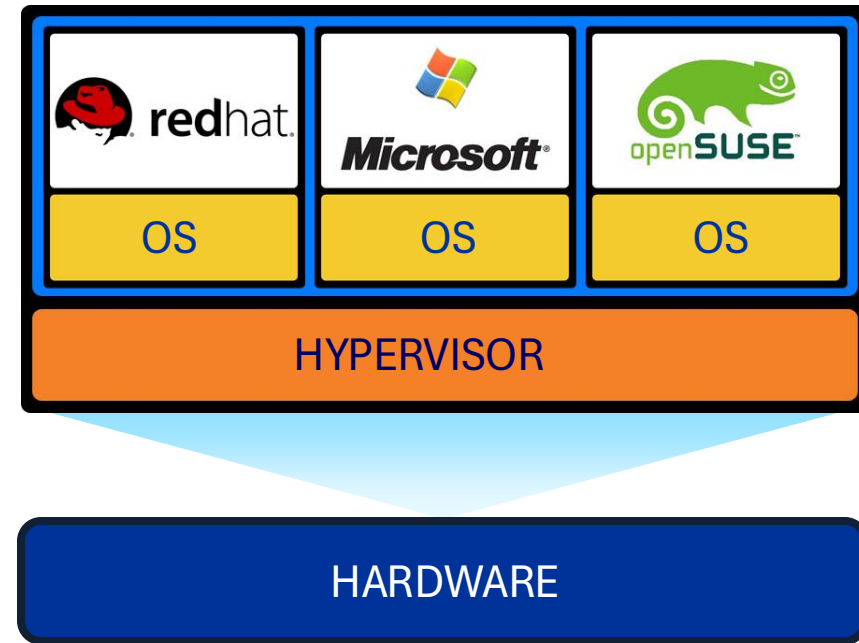
A Virtual Machine

- Virtualization consists of creating a virtual version of a resource that is normally provided physically and belongs to a system.
 - Better utilization of resources and computing units



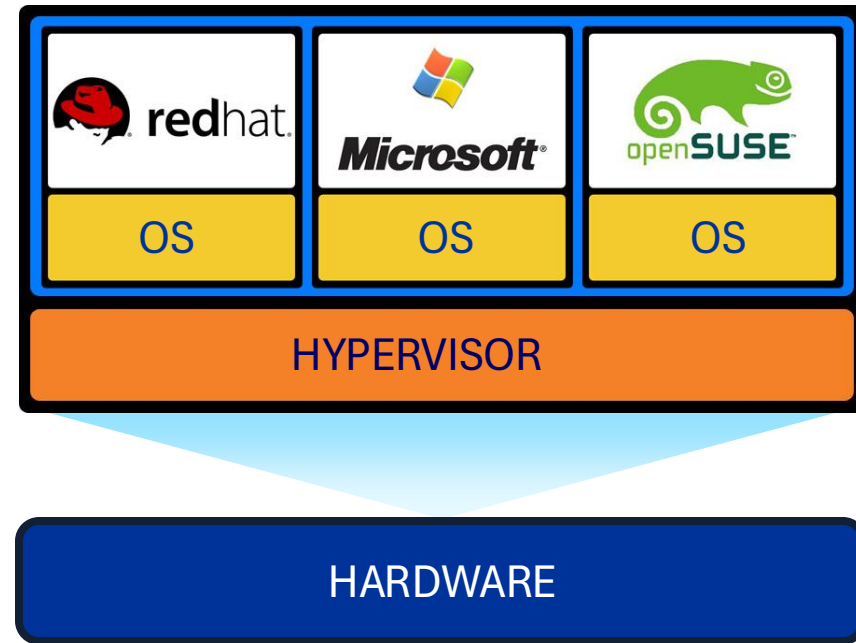
How is virtualization achieved?

- A new layer is introduced that runs before the operating system, called **Hypervisor**.
- The Hypervisor simulates the existence of multiple “virtual” computers on one physical computer.



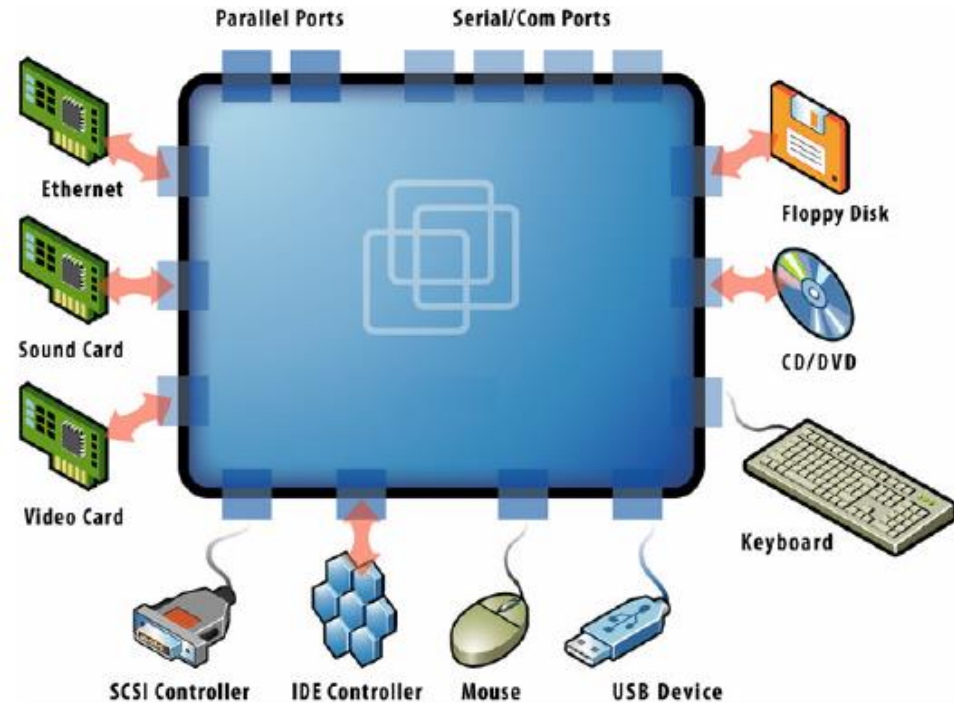
How is virtualization achieved?

- The hypervisor provides physical resources to virtualized computers and manages them concurrently for the benefit of virtual machines.
- A normal operating system is installed on each virtualized computer, which has no way of knowing whether it is running on a “real” or virtual computer.



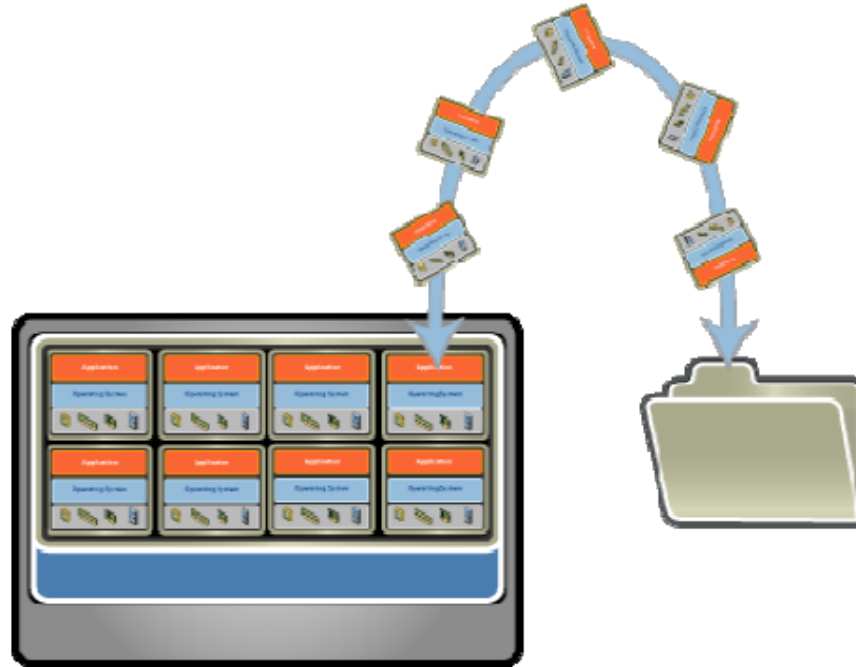
Hardware Independence

- The hypervisor simulates a standard server configuration, hiding the actual physical hardware.
- Consequence: Virtual machines can run on different physical servers without needing to be configured.



Storing a VM

- A virtual machine is **a file** containing:
 - Operating System, Applications, Data
 - Representation of Server Memory
- A file stores the state of a virtual server and allows it to be copied to any number of physical servers
- There are commercial tools that transform the contents of a physical server into a VM



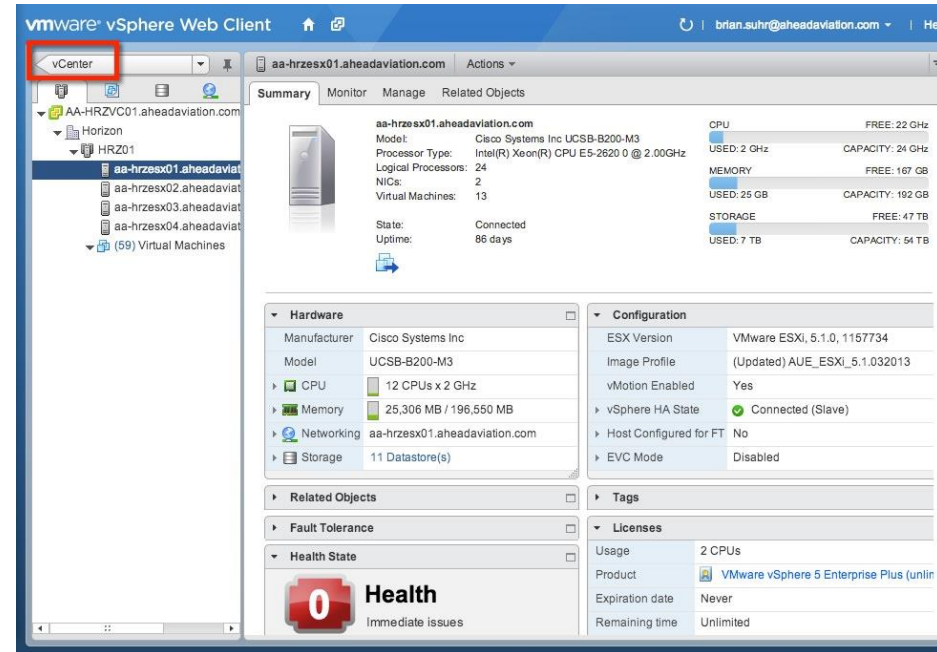
Isolating a VM

- Each VM is completely isolated from other VMs running on the same physical server
- In the event of an attack or crash on one VM, the others are not affected



VM management

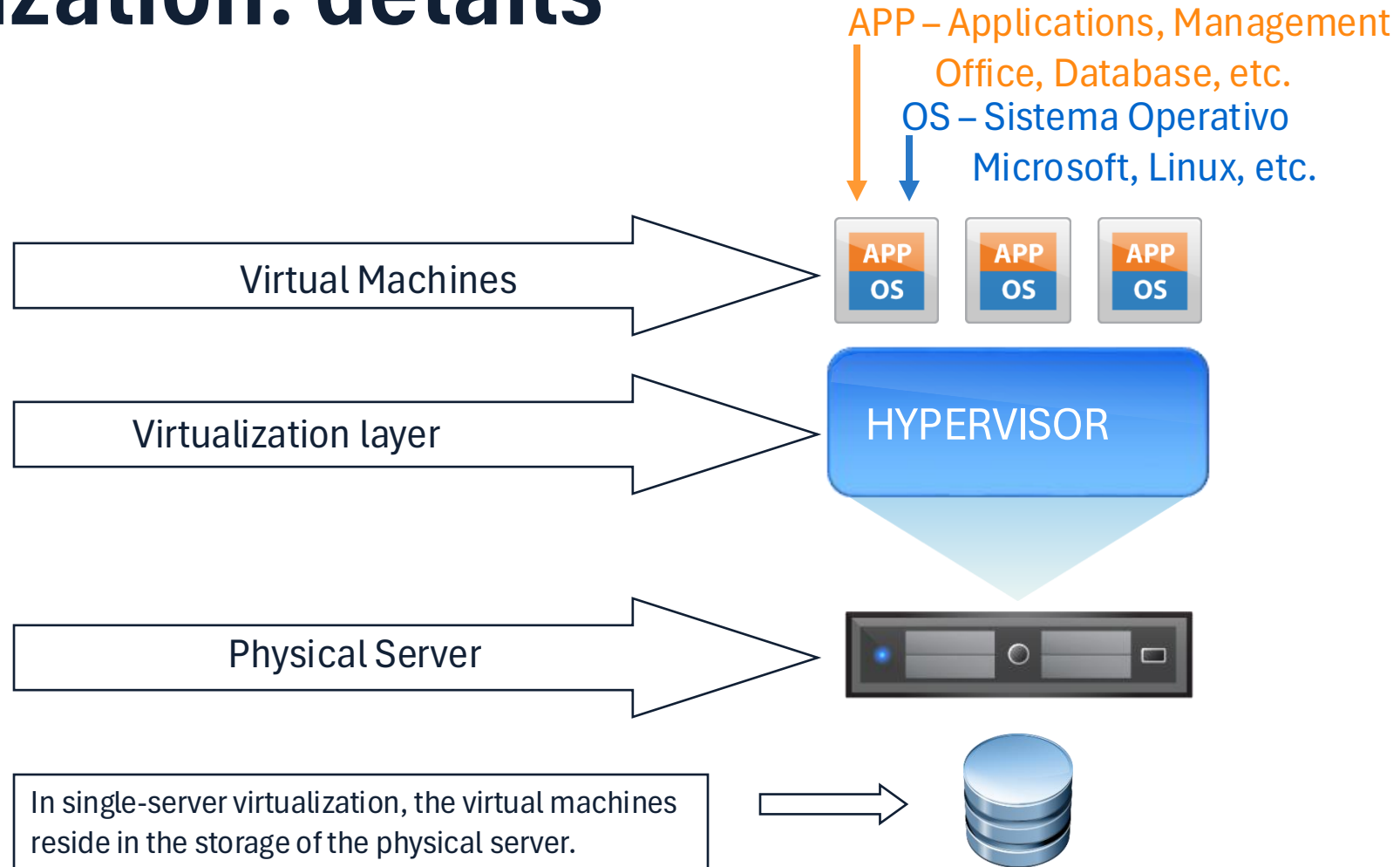
- All VMs in a data center can be managed through a single window.
- Management of the entire data center is centralized, greatly simplifying administration.



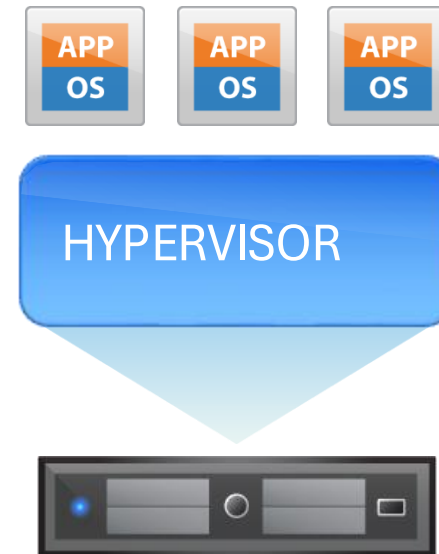
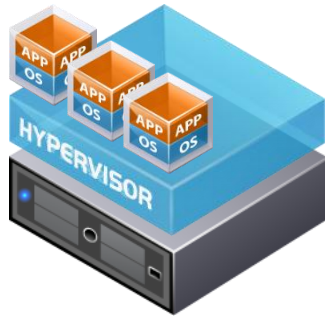
Virtual Machines

Operational Details

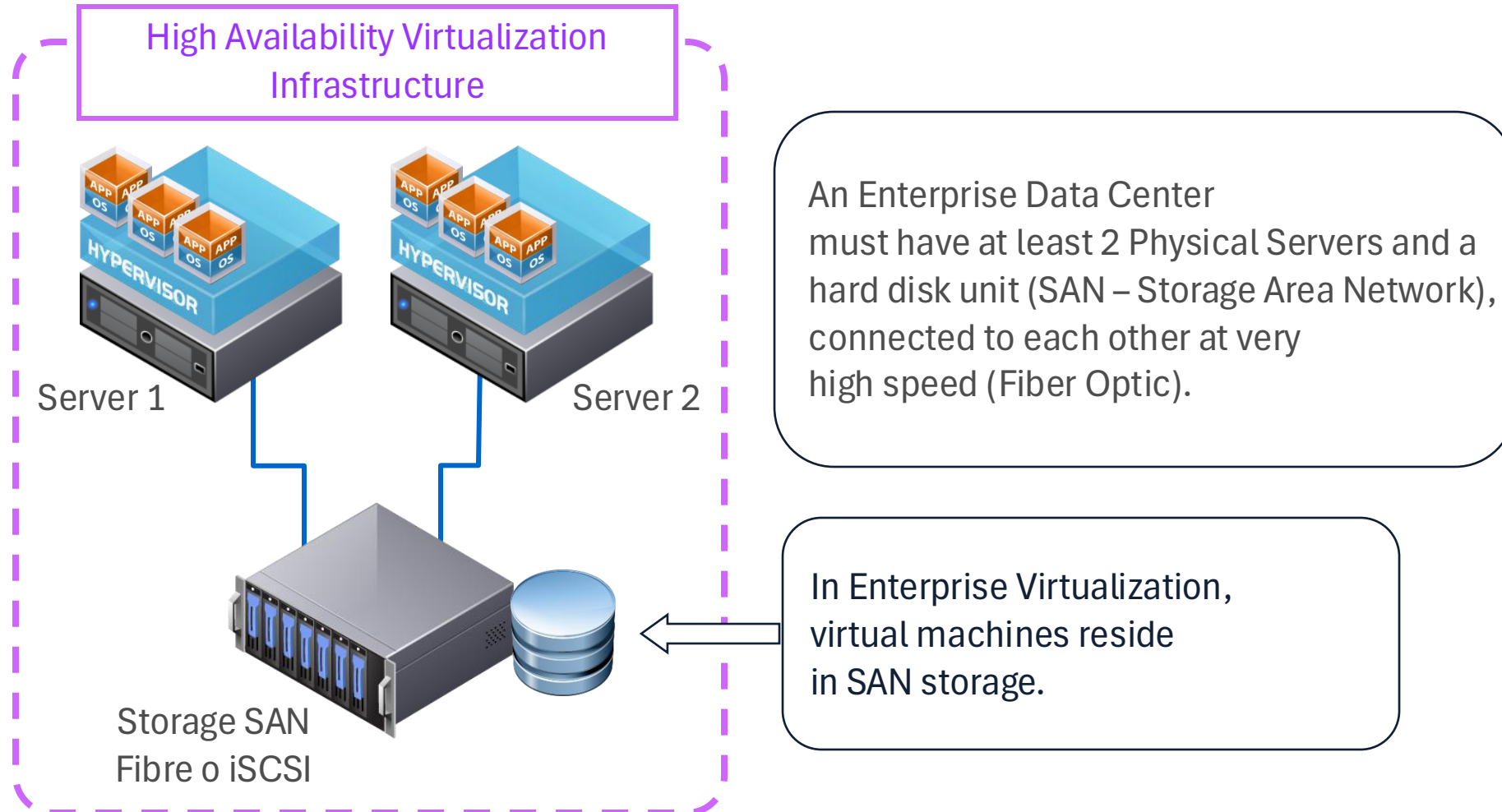
Virtualization: details



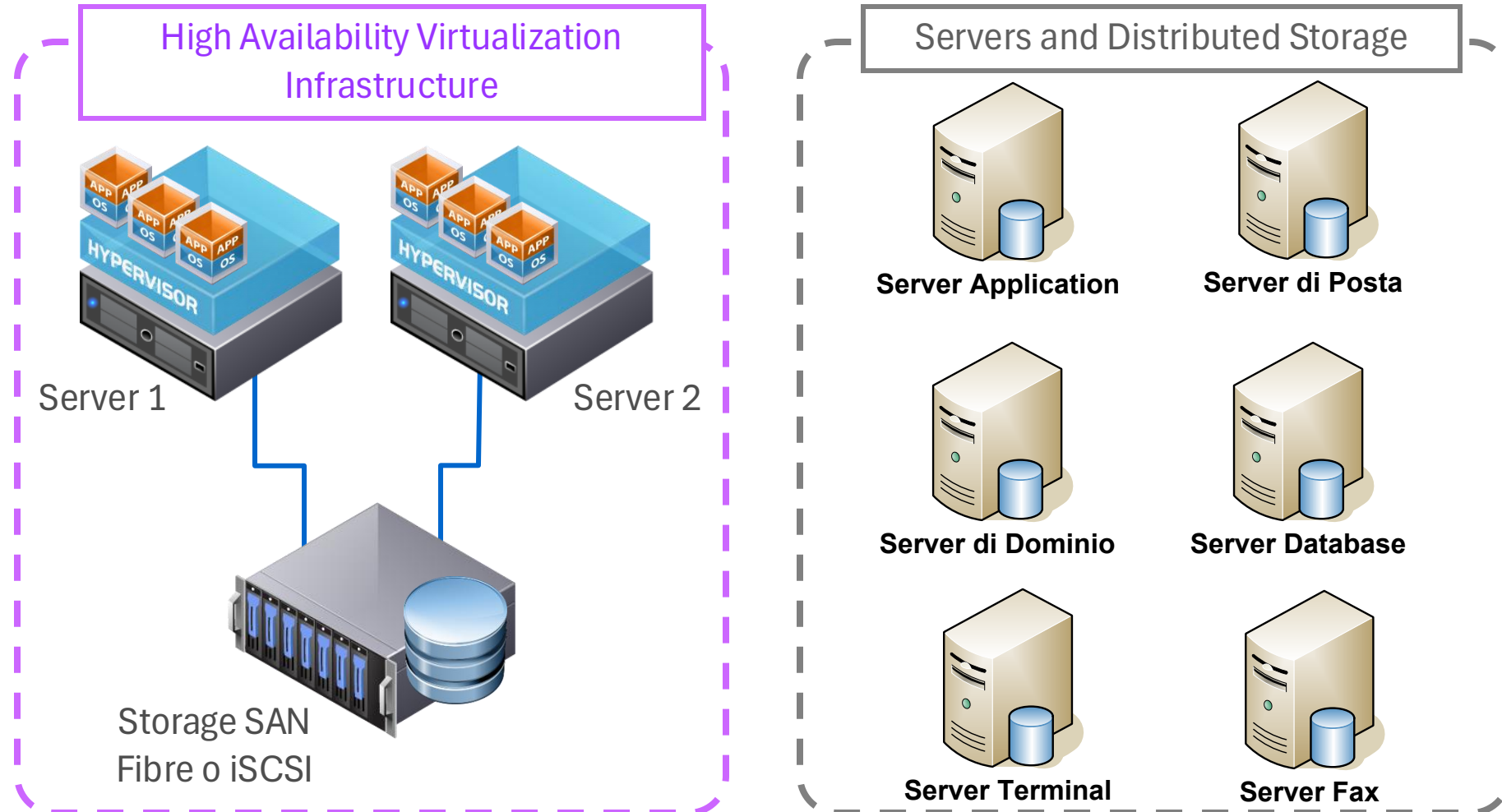
High Availability Enterprise Virtualization



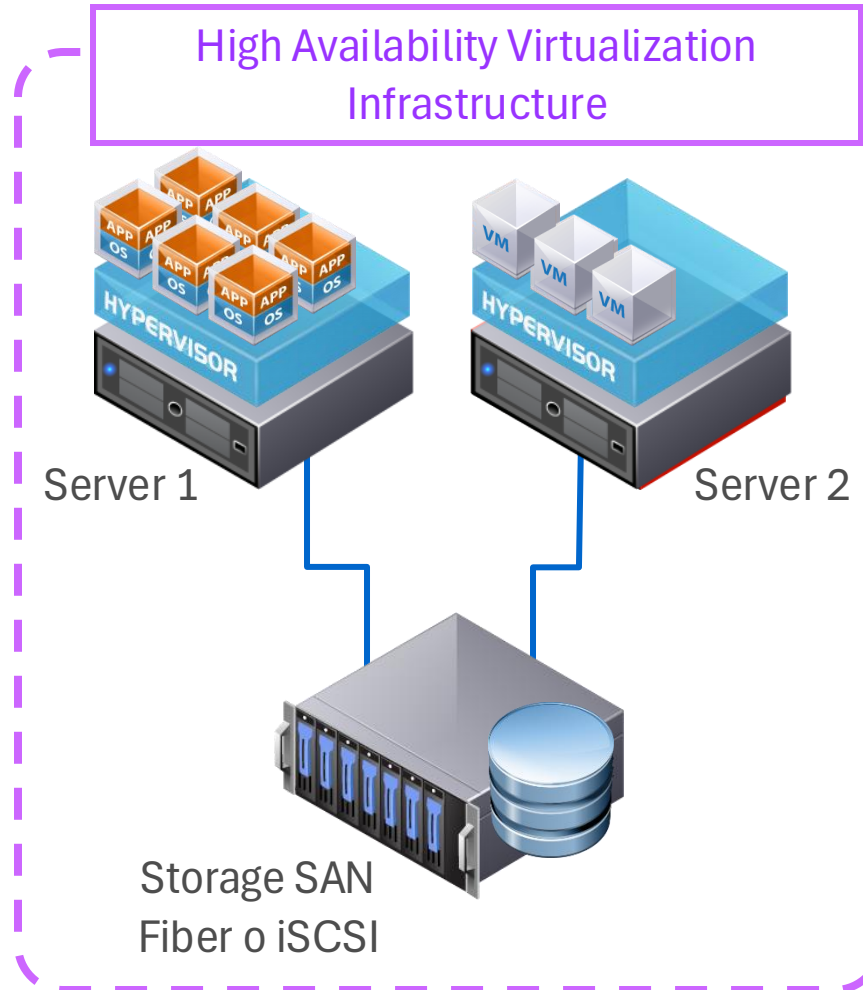
High Availability Enterprise Virtualization



Migration from Physical to Virtual



How a High Availability System Works



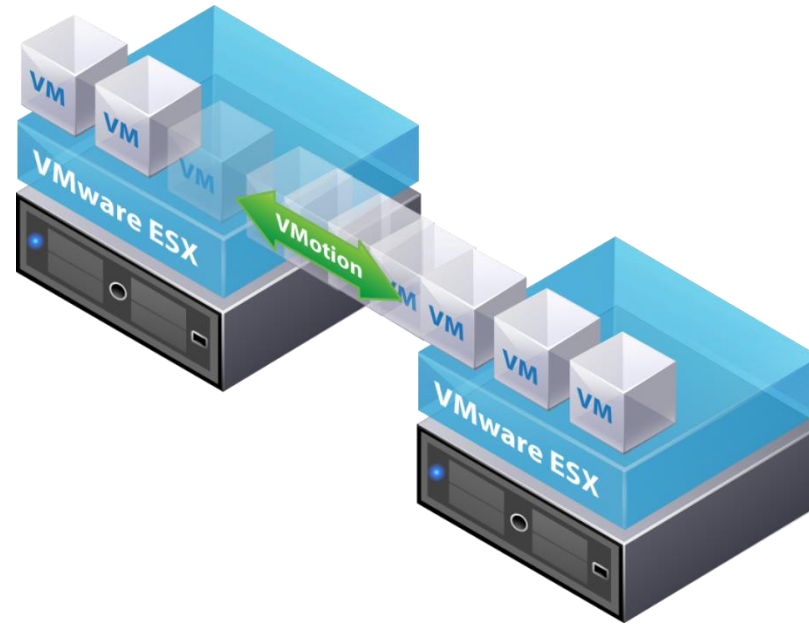
Virtualization ensures business continuity through the HA (High Availability) function.

If a server fails, the virtual machines on it will be instantly migrated to other available servers and restarted, minimizing downtime to the time it takes to restart the virtual machines.

Recent technologies also enable Fault Tolerance (FT), which guarantees the continuity of VM operation after a server failure, eliminating downtime. It is currently the best solution available for business continuity.

Virtualization: Live Migration of VMs

- The “hot” migration of VMs between different physical servers is one of the many features of the virtualization system.
- It allows maintenance or load balancing operations to be carried out while keeping the virtual servers operational.

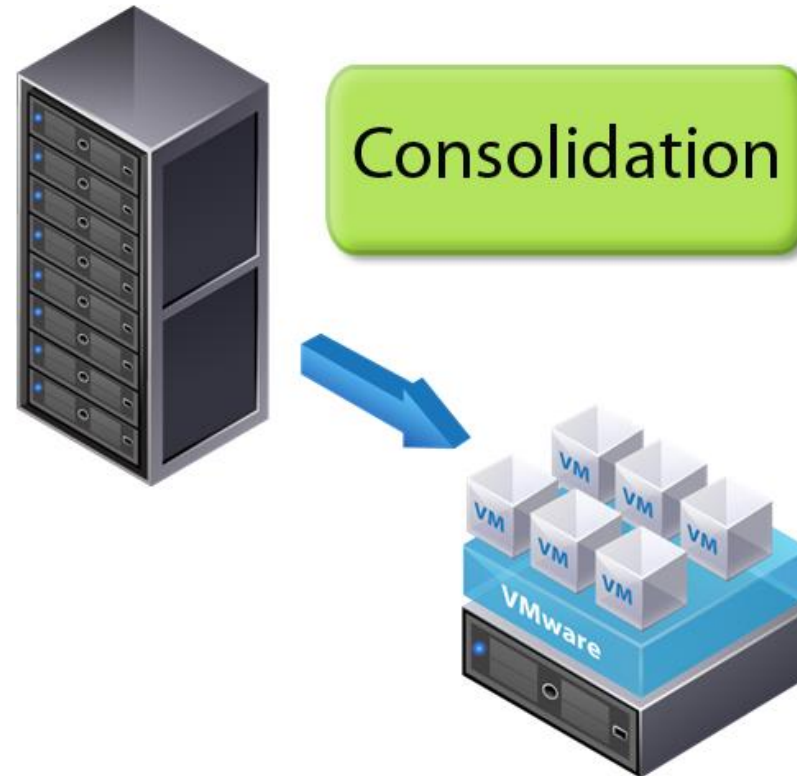


Virtual Machines

Benefits

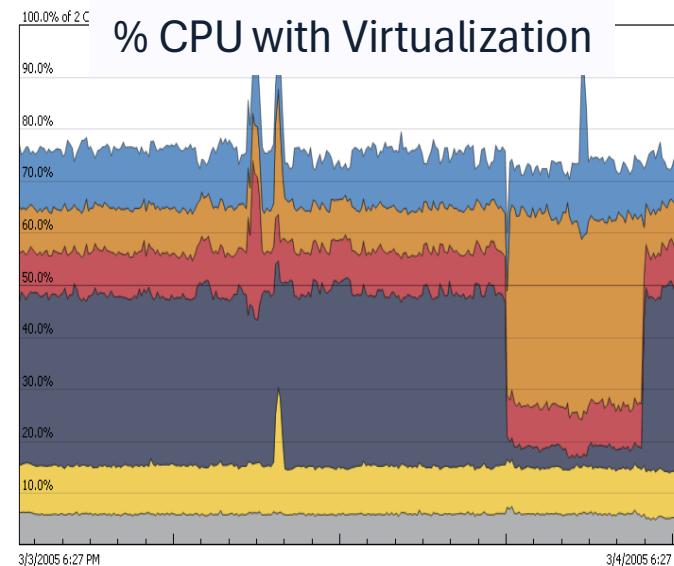
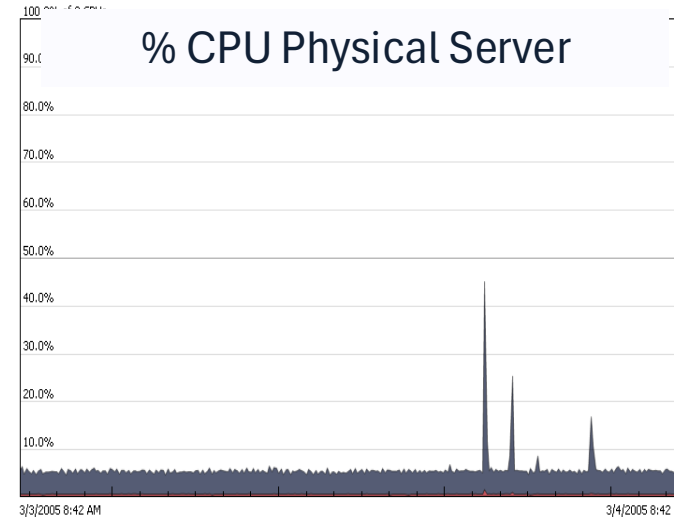
Consolidation of a data center

- With the introduction of virtualization, there is talk of “**consolidation**” of a data center.
- By making the most out of the parallel computing resources offered by new processors, it is possible to significantly reduce the number of servers needed to provide the IT services required by an organization.



Benefits introduced by virtualization: Optimization

- Optimization of hardware resources - Virtualization allows different operating systems to run simultaneously on a single machine.
- Optimization of disk space usage - A virtual machine is allocated only the disk space it strictly needs.
 - The rest of the volume remains available for new virtual machines.



Benefits introduced by virtualization:

Simplification

- Backup and disaster recovery
 - The entire server (OS, applications, data, devices, and status) is nothing more than a file, making it easier to create backup copies for disaster recovery.
- Hardware independence
- Simplified releases
 - Ability to clone virtual machines “LIVE” for testing and updates without compromising the integrity of the original.

Benefits introduced by virtualization:

Costs

- Reduction in IT costs
 - Reduction in the number of servers required, resulting in savings on energy and air conditioning costs and the physical space required by the data center.
- Reduction in IT personnel costs
 - Centralized management: Increased productivity of the IT department due to centralized management of machines through a single administration interface.

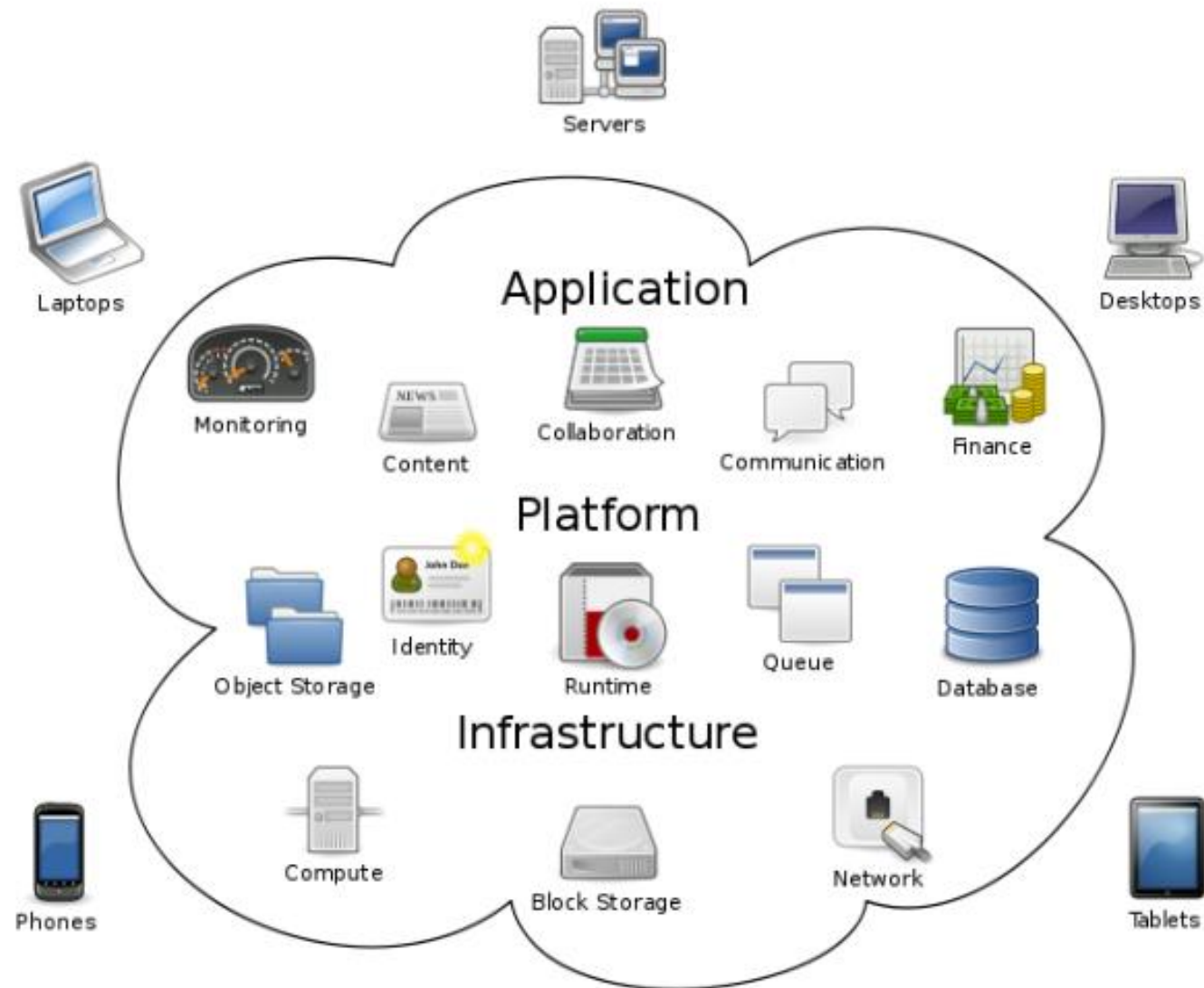
Example in a PA data center in Naples

- Approximately 800 virtual machines running in parallel, with load balancing.
- Approximately 30 physical servers.
- 2 Storage Area Networks, with approximately 500 TB of space.
- Management of 800 machines from 2 administration consoles.

Cloud Computing

Cloud Computing

- IT resources provided as a service
 - Compute, storage, databases, queues
- Clouds leverage economies of scale of commodity hardware
 - Cheap storage, high bandwidth networks & multicore processors
 - Geographically distributed data centers
- Offerings from Microsoft, Amazon, Google, ...



Cloud Computing

Cloud Scalability & Elasticity

Scalability & Elasticity

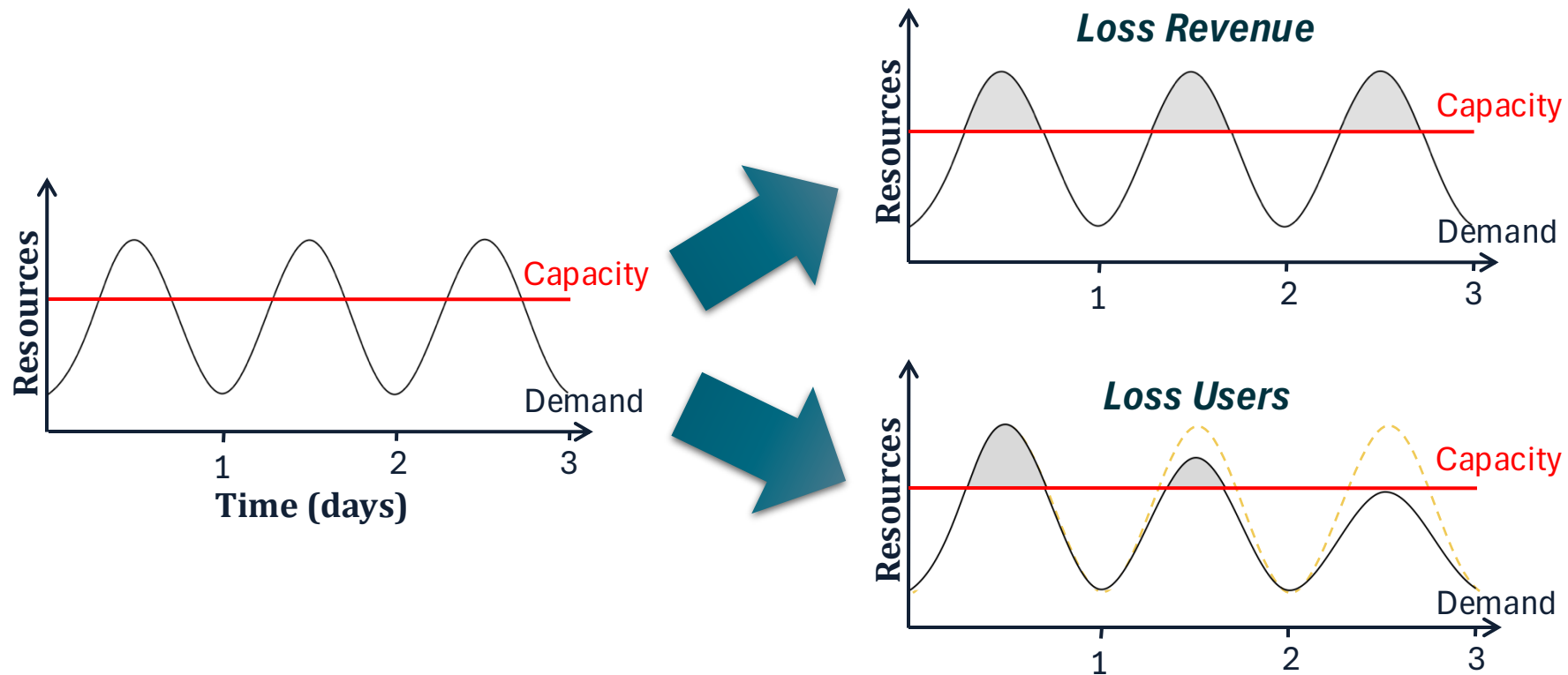
- What is scalability ?
 - A desirable property of a system, a network, or a process, which indicates its ability to either handle growing amounts of work in a graceful manner or to be readily enlarged.
- What is elasticity ?
 - The ability to grow or shrink infrastructure resources dynamically as needed to adapt to workload changes in an autonomic manner, maximizing the use of resources.
- But how to achieve these properties ?
 - Dynamic provisioning
 - Multi-tenant design

Dynamic Provisioning

- Dynamic Provisioning is a simplified way to explain a complex networked server computing environment where server computing instances are provisioned or deployed from a administrative console or client application by the server administrator, network administrator, or any other enabled user.

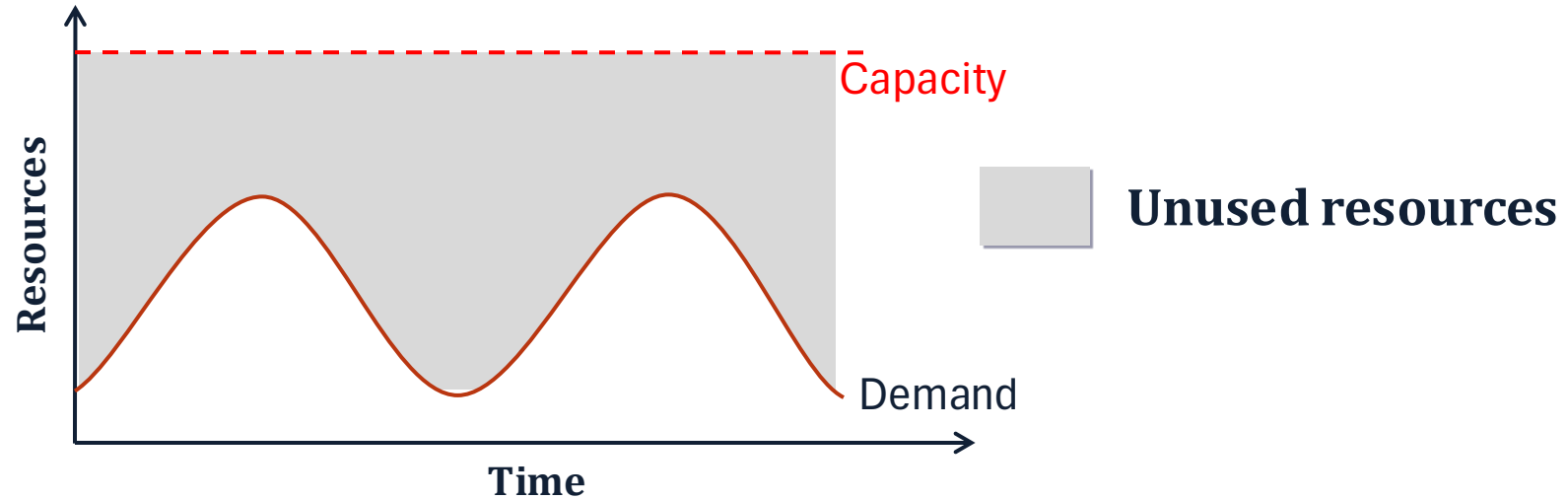
Dynamic Provisioning

- In traditional computing model, two common problems :
 - Underestimate system utilization which result in under provision



Dynamic Provisioning

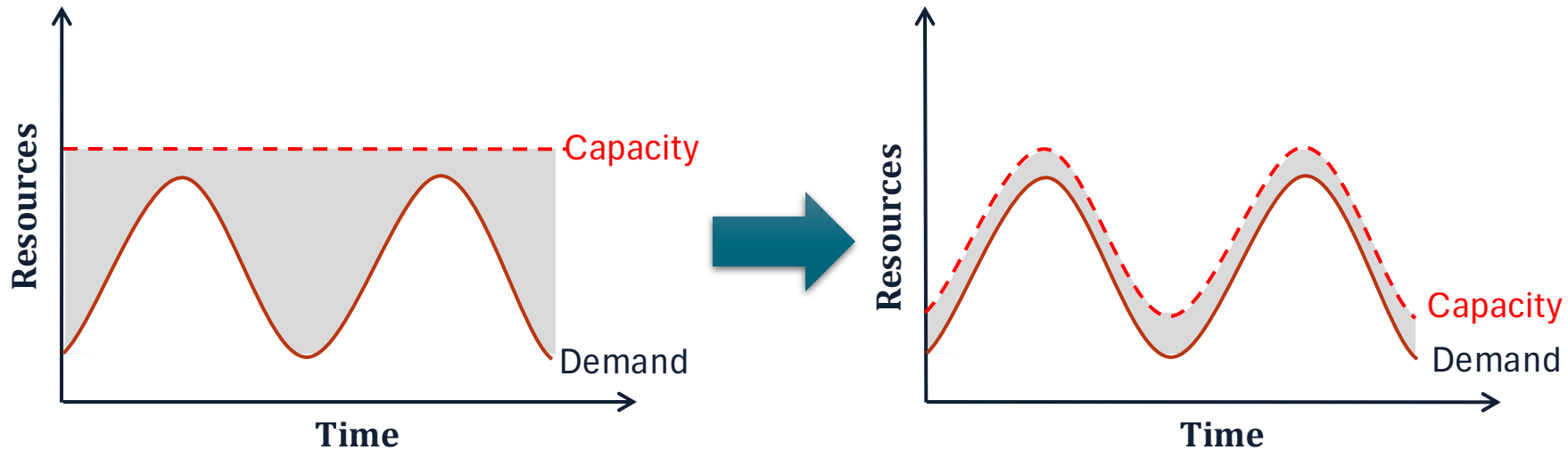
- Overestimate system utilization which result in low utilization



- How to solve this problem ??
 - Dynamically provision resources

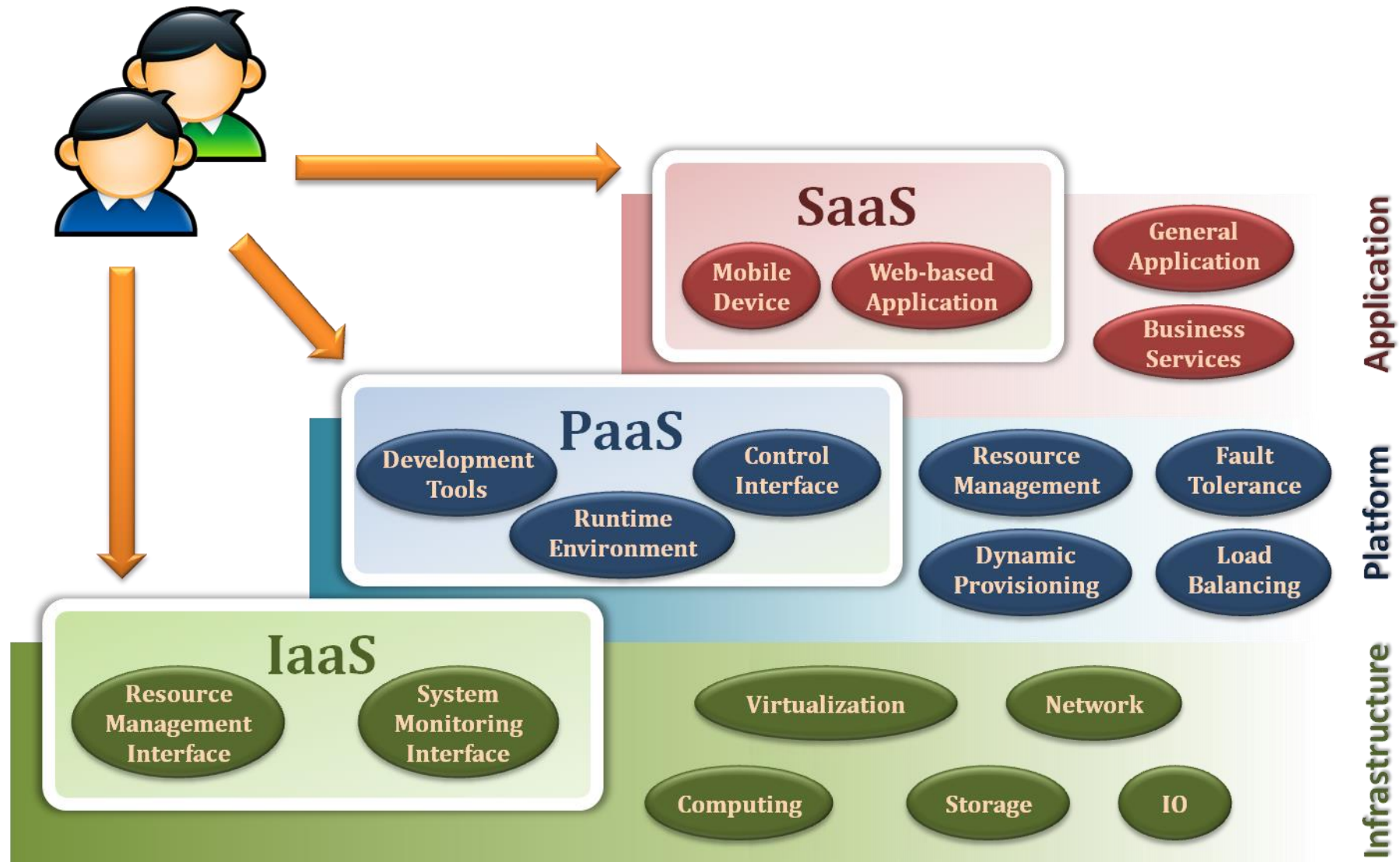
Dynamic Provisioning

- Cloud resources should be provisioned dynamically
 - Meet seasonal demand variations
 - Meet demand variations between different industries
 - Meet burst demand for some extraordinary events



Service Models

Service Model Overview

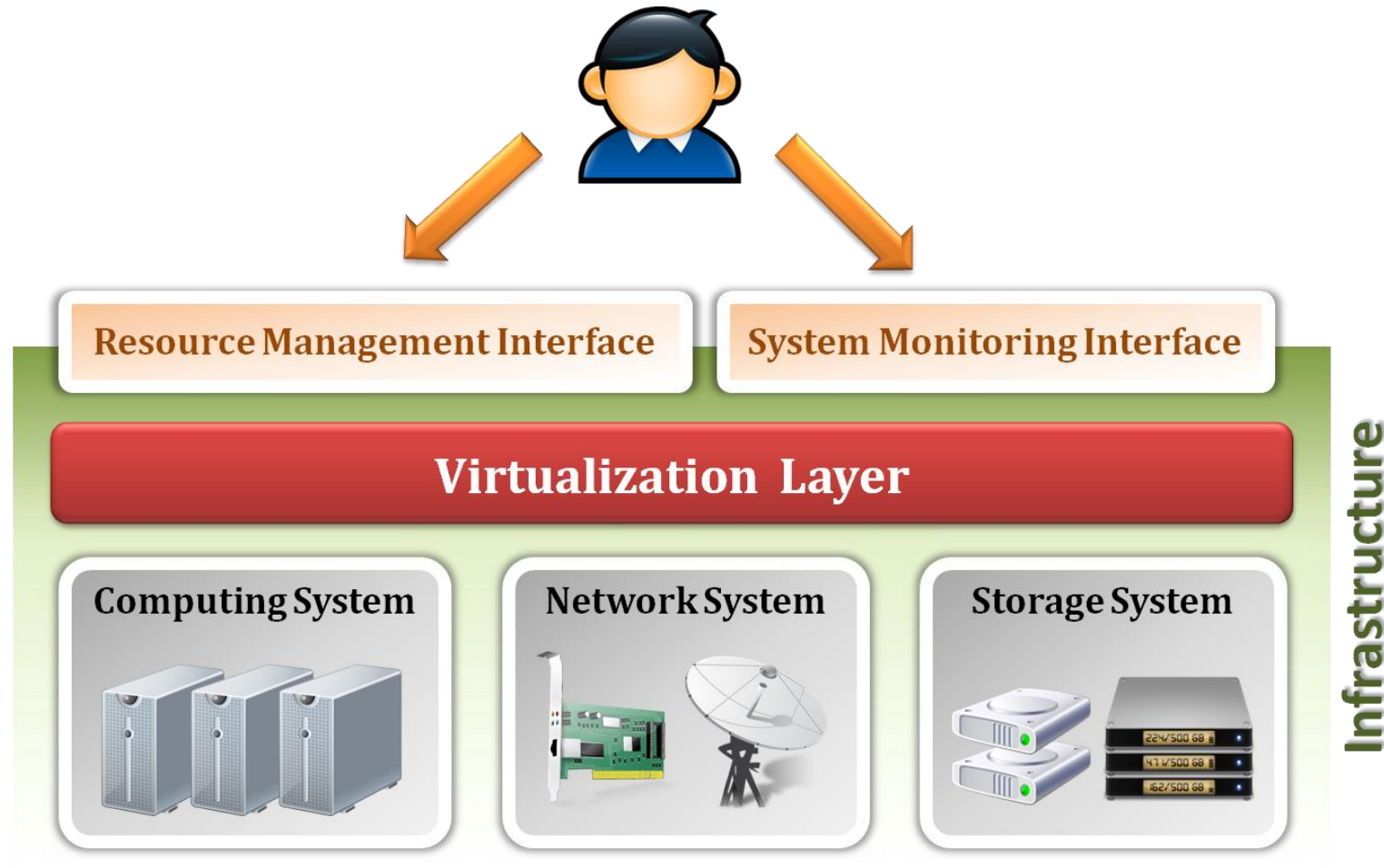


Infrastructure as a Service - IaaS

- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
 - The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components .
- Examples :
 - Amazon EC2
 - Eucalyputs
 - OpenNebula
 - ... etc

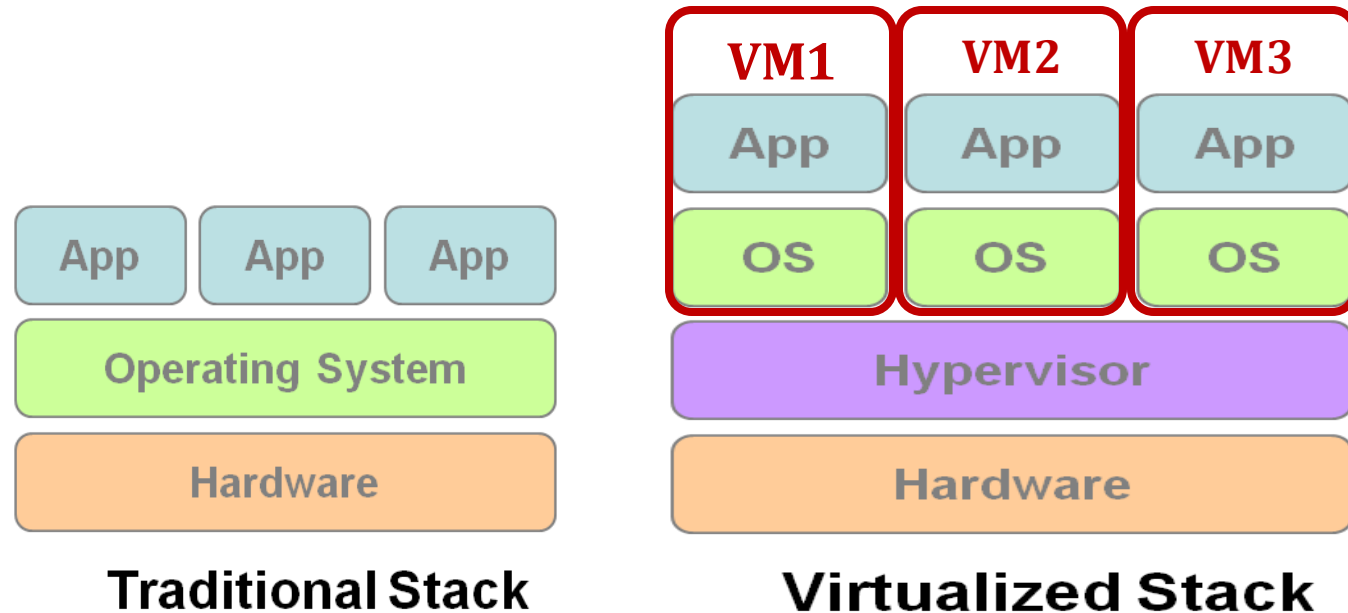
Infrastructure as a Service

- System architecture :



Infrastructure as a Service

- Enabling technique - Virtualization
 - Virtualization is an abstraction of logical resources away from underlying physical resources.
 - Virtualization technique shift OS onto hypervisor.
 - Multiple OS share the physical hardware and provide different services.
 - Improve utilization, availability, security and convenience.

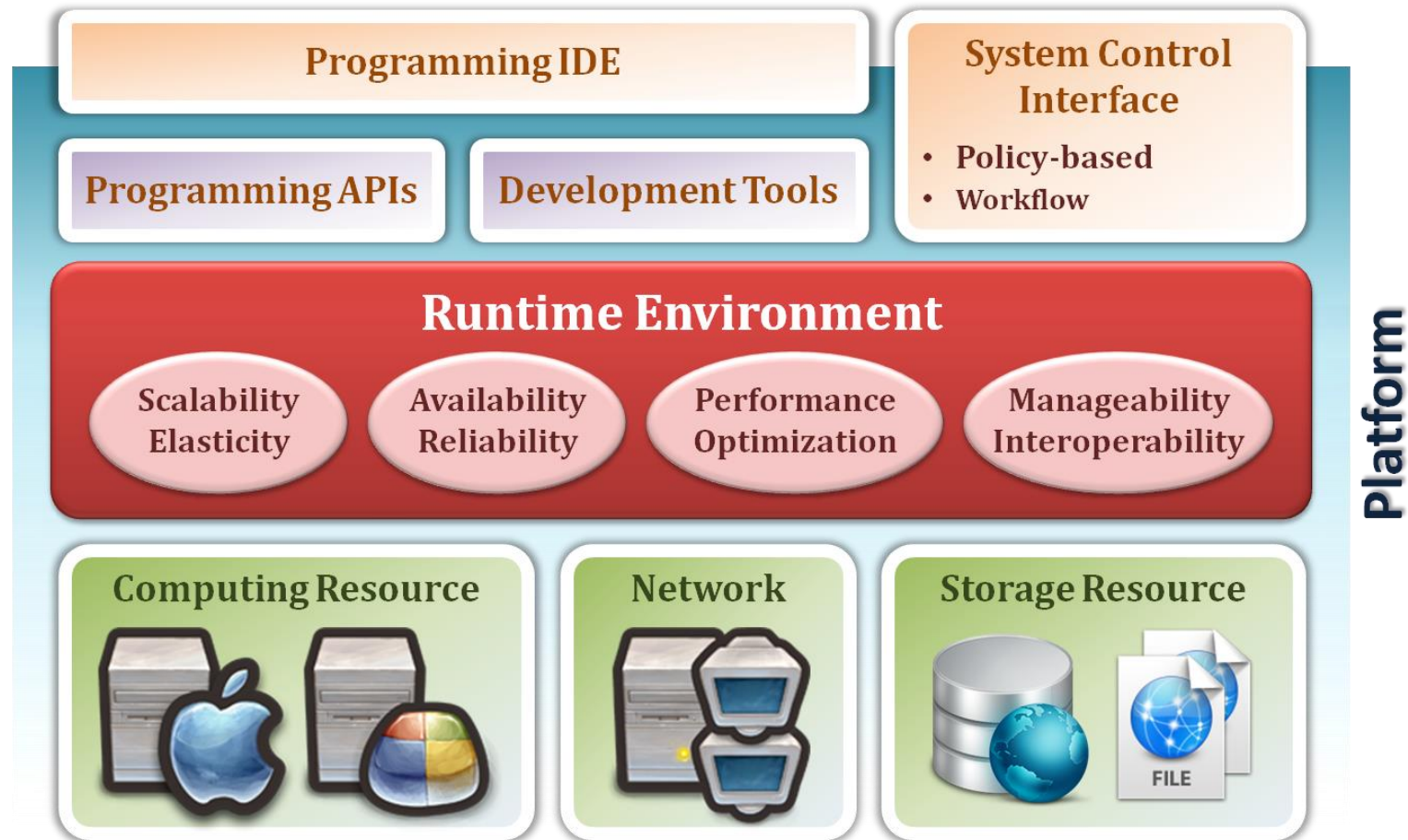


Platform as a Service - PaaS

- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider.
 - The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.
- Examples :
 - Microsoft Windows Azure
 - Google App Engine
 - Hadoop
 - ... etc

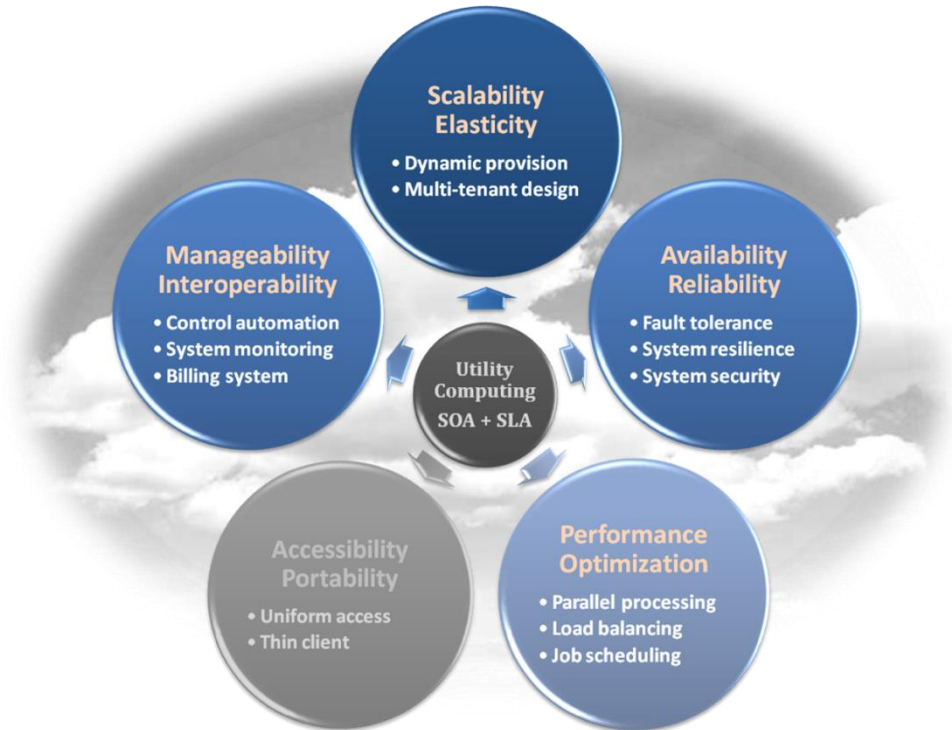
Platform as a Service

- System architecture :



Platform as a Service

- Enabling technique – Runtime Environment Design
 - Runtime environment refers to collection of software services available. Usually implemented by a collection of program libraries.
- Common properties in Runtime Environment :
 - Manageability and Interoperability
 - Performance and Optimization
 - Availability and Reliability
 - Scalability and Elasticity



Platform as a Service

- Provide service – **Programming IDE**
 - Users make use of programming IDE to develop their service among PaaS.
 - This IDE should integrate the full functionalities which supported from the underlying runtime environment.
 - This IDE should also provide some development tools, such as profiler, debugger and testing environment.
 - The programming APIs supported from runtime environment may be various between different cloud providers, but there are still some common operating functions.
 - Computation, storage and communication resource operation

Software as a Service - SaaS

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email).
 - The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.
- Examples :
 - Google Apps (e.g., Gmail, Google Docs, Google sites, ...etc)
 - Salesforce.com
 - EyeOS
 - ... etc

Software as a Service



Software as a Service

- Provide service – Web-based Applications
 - Conventional applications should translate their access interface onto web-based platform.
 - Applications in different domains
 - General Applications – Applications which are designed for general propose, such as office suit, multimedia and instant message, ...etc.
 - Business Applications – Application which are designed for business propose, such as ERP, CRM and market trading system, ...etc.
 - Scientific Applications – Application which are designed for scientific propose, such as aerospace simulation and biochemistry simulation, ...etc.
 - Government Applications – Applications which are designed for government propose, such as national medical system and public transportation system service, ...etc.

Deployment models

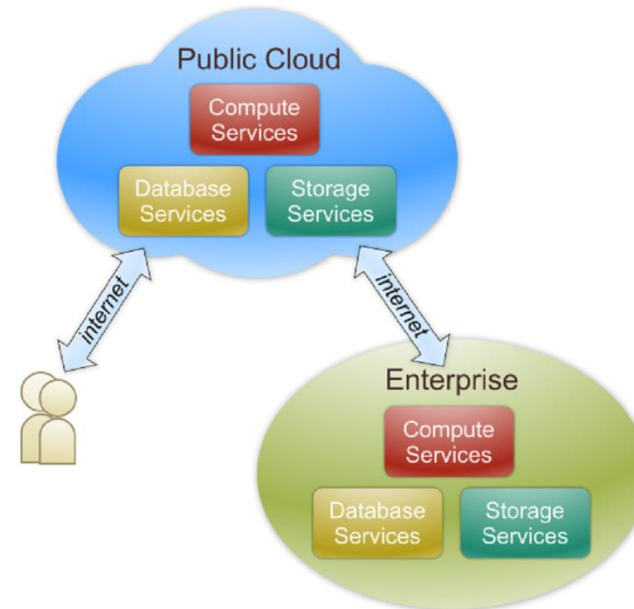
How to deploy a cloud system ?

Deployment Model

- There are three primary cloud deployment models :
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud
- Each can exhibit the previously discussed characteristics; their differences lie primarily in the scope and access of published cloud services, as they are made available to service consumers.

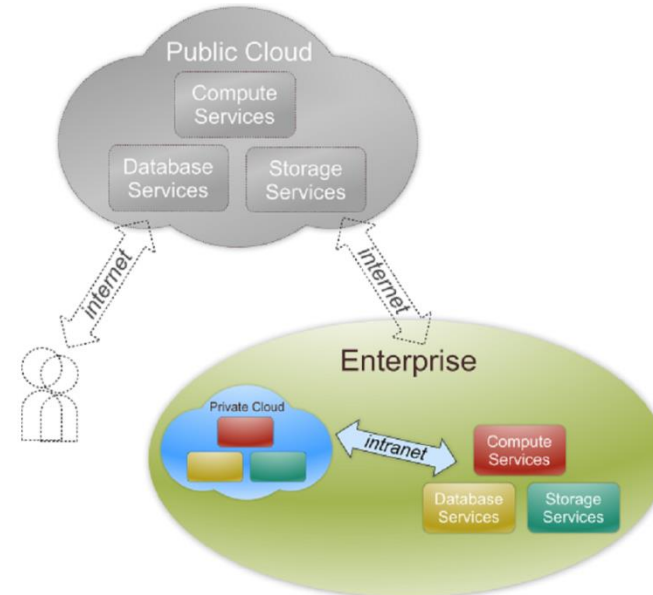
Public Cloud

- The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.
 - Also known as external cloud or multi-tenant cloud, this model essentially represents a cloud environment that is openly accessible.
 - Basic characteristics :
 - Homogeneous infrastructure
 - Common policies
 - Shared resources and multi-tenant
 - Leased or rented infrastructure
 - Economies of scale



Private Cloud

- The cloud infrastructure is operated solely for an organization. It may be managed by the organization or a third party and may exist on premise or off premise.
 - Also referred to as internal cloud or on-premise cloud, a private cloud intentionally limits access to its resources to service consumers that belong to the same organization that owns the cloud.
- Basic characteristics :
 - Heterogeneous infrastructure
 - Customized and tailored policies
 - Dedicated resources
 - In-house infrastructure
 - End-to-end control



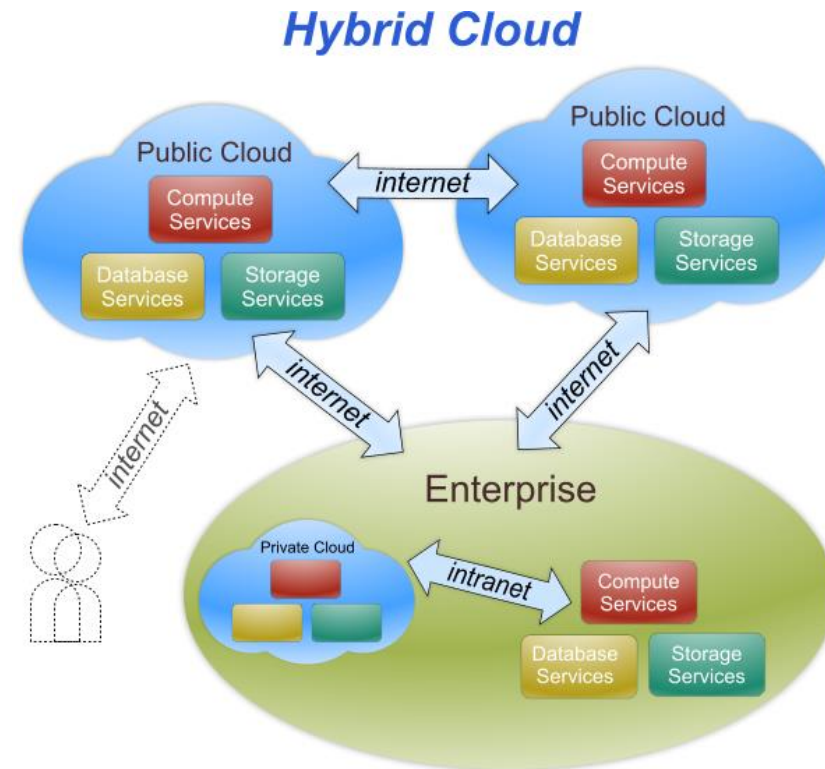
Public vs. Private

- Comparison :

	Public Cloud	Private Cloud
<i>Infrastructure</i>	<i>Homogeneous</i>	<i>Heterogeneous</i>
<i>Policy Model</i>	<i>Common defined</i>	<i>Customized & Tailored</i>
<i>Resource Model</i>	<i>Shared & Multi-tenant</i>	<i>Dedicated</i>
<i>Cost Model</i>	<i>Operational expenditure</i>	<i>Capital expenditure</i>
<i>Economy Model</i>	<i>Large economy of scale</i>	<i>End-to-end control</i>

Hybrid Cloud

- The cloud infrastructure is a composition of two or more clouds (private or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).



Cloud Ecosystem

